Counting and double counting

Notes from Carlos' lessons

Source lessons: 2026-02-11 (proving bijections); pigeonhole fragment at the left of 2026-02-27 (Ramsey numbers).

Iverson bracket

For a proposition P, $[P] = 1$ if P is true and 0 otherwise. Rules (check by truth table):

1. $[P \wedge Q] = [P][Q]$
2. $[P \vee Q] = [P] + [Q] - [P][Q]$
3. $[\neg P] = 1 - [P]$

Cardinality as a sum: for finite X and $S \subseteq X$, $|S| = \sum_{x \in X} [x \in S]$. Hence $n - |S| = \sum_{t \in [n]} (1 - [t \in S]) = \sum_{t \in [n]} [t \notin S]$ (used below).

The technique of the whole lesson: write a count as a double sum of brackets, swap the order of summation, and evaluate the other way.

Handshake lemma

Theorem 1 (Handshake lemma)**.** *For a finite undirected graph $G = (V, E)$ (edges are 2-element subsets of V),* $\sum_{v \in V} \deg(v) = 2|E|$.

Proof. $\deg(v) = |\{e \in E : v \in e\}| = \sum_{e \in E} [v \in e]$. So

$$\sum_v \deg(v) = \sum_v \sum_e [v \in e] = \sum_e \sum_v [v \in e] = \sum_e |\{v : v \in e\}| = \sum_e 2 = 2|E|.$$

The swap is legal because both sums are finite. The set being double counted is $\{(v, e) \in V \times E : v \in e\}$. □

Corollary 2. *The number of odd-degree vertices is even; in particular a graph on an odd number of vertices cannot have all degrees odd.*

(This is used in the parity argument for $R(3, 4) \leq 9$.)

$\sum_k \binom{n}{k} = 2^n$

Theorem 3. $\displaystyle\sum_{k=0}^{n} \binom{n}{k} = 2^n.$

Proof. $\binom{n}{k} = |\{S \subseteq [n] : |S| = k\}| = \sum_{S\subseteq[n]} [|S| = k]$. Sum over $k = 0, \ldots, n$ and swap:

$$\sum_{k=0}^{n} \binom{n}{k} = \sum_{S\subseteq[n]} \sum_{k=0}^{n} [|S| = k] = \sum_{S\subseteq[n]} 1 = |\mathcal{P}([n])| = 2^n,$$

since for each S exactly one k (namely $|S|$) makes the bracket equal to 1. □

Binomial theorem

Theorem 4. $(x+y)^n = \displaystyle\sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$

Proof. Expand the product $\prod_{t=1}^{n}(x+y)$. Choosing x or y from each factor t corresponds to choosing the set $S \subseteq [n]$ of factors that contribute x. So

$$(x+y)^n = \sum_{S\subseteq[n]} \prod_{t\in[n]} x^{[t\in S]} y^{[t\notin S]} = \sum_{S\subseteq[n]} x^{|S|} y^{n-|S|}.$$

Group the sets S by size $k = |S|$; there are $\binom{n}{k}$ of each size. □

The board example: $n = 3$, $S = \{1, 3\}$ picks $x \cdot y \cdot x$. A second proof, via exponential generating functions, is in the topic *Exponential generating functions.*

Pigeonhole: a block of terms divisible by N

The left margin of the 02-27 board asks: given $a_1, \ldots, a_N \in \mathbb{N}$, is there a nonempty S with $N \mid \sum_{k\in S} a_k$? It lists the prefix sums $a_1,\ a_1 + a_2,\ \ldots,\ a_1 + \cdots + a_N$, which is the standard proof.

Claim 5. *Some nonempty block of* consecutive *terms has sum divisible by N.*

Proof. Let $s_j \equiv a_1 + \cdots + a_j \pmod N$ for $j = 1, \ldots, N$. If some $s_j = 0$, the block $a_1, \ldots, a_j$ works. Otherwise the N values lie in $\{1, \ldots, N-1\}$, so two coincide, $s_i = s_j$ with $i < j$, and then $a_{i+1} + \cdots + a_j \equiv 0 \pmod N$. □

Notes

- The lesson's filename says "bijections", but what is on the board is the Iverson-bracket / double-counting toolkit; no explicit bijection is proved there.
- The right-hand scrawl on the board (the $\prod x^{[t\in S]} y^{[t\notin S]}$ expansion and the $S = \{1, 3\}$ example) is the binomial-theorem proof above.